

Project Initialization and Planning Phase

Date	10 july 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

The proposed solution introduces an AI-powered Emotion-Aware Learning Support Engine (EALSE) that evaluates a learner's emotional state through text processing. The system converts student messages into emotional classifications using BiLSTM and DistilBERT, and provides tailored guidance using Google Gemini AI, along with comprehensive analytics.

Project Overview

Topic	Description
Objective	Develop an intelligent application that identifies student emotional states and provides empathetic, AI-generated guidance.
Scope	The system supports text input, BiLSTM & DistilBERT emotion classification, Gemini AI guidance generation, and interaction logging.

Problem Statement

Topic	Description
Description	Traditional AI tutors deliver uniform responses regardless of the learner's underlying emotional state, ignoring frustration or confusion.
Impact	This misalignment reduces engagement, worsens learning outcomes, and leaves educators without actionable emotional insights.

Proposed Solution

Topic	Details
Approach	Use BiLSTM, DistilBERT, NLTK preprocessing, and Gemini AI to evaluate emotions and generate support.

Key Features	• Text emotion classification • Mixed emotion detection • Ensemble model inference • Role-aware Gemini guidance • Interactive dashboard
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Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Processor	CPU	Intel Core i3 / Ryzen 3 or higher
Memory	RAM	4 GB or above
Storage	Disk Space	2 GB free space
Software		
Framework	Application Framework	Streamlit
Libraries	Python Libraries	TensorFlow, PyTorch, HuggingFace, NLTK, Google GenAI, Pl
Development	IDE	Visual Studio Code
Data		
Dataset	GoEmotions	CSV / HF Dataset